

Michelle Zhuang

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EDUCATION

University of Washington, Seattle, WA

Expected May 2027

MS in Artificial Intelligence and Machine Learning for Engineering

- Relevant Coursework: Computer Vision and Deep Learning, Natural Language Processing; implemented transformers/self-attention from scratch, SimCLR, LSTMs, and RL (Q-learning, PPO)

Swarthmore College, Swarthmore, PA

May 2023

BS Engineering and BA Computer Science Double Major – GPA: 3.8/4.0

- Relevant Coursework: Data Structures and Algorithms, Computer Architecture, Computer Systems, Embedded Systems, Computer Vision, Mobile Robotics, Game Systems, Machine Learning
- Awards: Helen North Scholarship, Frances Velay Women's Science Research Fellowship, Grace Hopper Scholar

TECHNICAL SKILLS

ML/AI & Data: PyTorch, TensorFlow, Hugging Face Transformers, LoRA/PEFT, OpenCV, LangChain, LangGraph, NumPy, Pandas, Matplotlib, R

Programming & Development: Python, Java, C++, JavaScript, SQL, PL/SQL, Git/GitHub, Docker, Kubernetes, Unix

Platforms & Tools: AWS, Railway, WorkOS, FastAPI, Flask, PostgreSQL, SQLAlchemy, React, Angular, HTML/CSS

Certificates: Google IT Automation with Python, Google Prompting Essentials

WORK EXPERIENCE

Software Engineer, Cisco, Research Triangle Park, NC

July 2023 – November 2024

- Developed PL/SQL packages automating ingestion of leasing data, improving data integration efficiency by 20%
- Built and deployed automated testing workflows using UiPath, automating 60 test cases and improving deployment reliability across environments
- Collaborated cross-functionally with Finance, Operations, and Accounting teams to deliver scalable data solutions supporting \$11.8 billion bookings

Software Engineer Intern, Oracle OCI, Remote

May 2022 – August 2022

- Implemented data pipelining program using Apache Kafka APIs in Docker containers, processing over 10,000 security findings daily
- Built visualization dashboards in OpenSearch to provide real-time monitoring of security metrics

PROJECTS

Fitness & Nutrition Tracker

August 2026 – Present

- Built a full-stack fitness and nutrition tracker (FastAPI, PostgreSQL/SQLAlchemy, React) with a dual-transport backend, letting a web client and an AI agent share one service and authorization layer instead of duplicating logic
- Built an MCP server exposing 30+ tools (workout logging, templates, nutrition lookup) so an AI agent operates the app through the same service layer as the web client; secured both with WorkOS AuthKit
- Diagnosed and fixed a database race condition causing duplicate race conditions and a staging/production token-refresh bug; deployed on Railway and used daily

Revisiting RecipeGPT

June 2026 – August 2026

- Fine-tuned GPT-2 for recipe generation and benchmarked LoRA against full fine-tuning across a rank sweep; LoRA at rank 4 came within 99.4% of full fine-tuning on BERTScore-F1 while training only 0.24% of parameters

ThriftFit: Virtual Try-On Robustness Study

May 2026 – July 2026

- Evaluated the robustness of OOTDiffusion virtual try-on across 200 garment and pose conditions, pairing qualitative failure-mode scoring with LPIPS/SSIM metrics

Feature Tracing and Projection, Swarthmore College, Swarthmore, PA

October 2022 – May 2023

- Implemented real-time computer vision system using Python and OpenCV to correct fabric deformation for accurate sewing pattern projections
- Developed algorithm to maintain calibration between camera input and projector output, enabling precise pattern mapping despite fabric movement

Computer Vision, Swarthmore College, Swarthmore, PA

January 2022 – May 2022

- Built live face swap application using OpenCV and Laplacian blurring, achieving real-time processing

Computational Sewing Pattern Generator, Swarthmore College, Swarthmore, PA

May 2021 – August 2021

- Designed algorithmic pattern-making system integrated with Autodesk Fusion 360, automating 3D-to-2D garment pattern conversion
- Created parametric design system for allowing customization based on measurements

LEADERSHIP EXPERIENCE

Club President, Women in Computer Science (W+iCS), Swarthmore, PA

January 2021 – May 2023

- Organized 10+ networking events and research panels, increasing membership engagement by 30%
- Maintained club website and digital presence, implementing new features to improve user engagement